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SpaceCDF Instructor Book

Supplement to the SpaceCDF Course Textbook

SpaceCDF
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Faculty of Engineering · School of Engineering Design and Teaching Innovation (SEDTI)

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Classification: Internal

Supplement to SpaceCDF_Textbook.pdf · Audience: simulator operator and instructor coach

This short volume covers the simulator-side material referenced in the textbook's Part 2 (CubeSat Operations). It is not for trainees on console; it is the runbook for the person driving the simulator and the coach watching the room.

Contents:

1. Simulator setup and breakpoints
2. Anomaly scenario catalogue (per-scenario YAML, cue moments, expected operator response)
3. Multi-anomaly composition
4. Coaching protocols and pass-criteria scoring

Chapter 5.10 — Scenario Catalogue

This chapter is the instructor's anomaly-injection menu. For each scenario, the table gives the YAML filename, scenario name, difficulty, sim duration, the canonical recovery procedure, and the recommended cue moment (when in a pass to inject it).

5.10.0 Starting the simulator

The simulator takes `--config` (default `configs/eosat1/`), `--speed` (default 1.0), `--host` and `--port`. `--port` defaults to `$SMO_SIM_HTTP_PORT` and then to **8080**; the MCS and the planner take their own `--port`, so if 8080 is already in use on the teaching host, move the simulator rather than the clients:

```
smo-simulator --config configs/eosat1/ --port 8081 --speed 1.0
```

Two operating rules that shape everything below:

- **Phase transitions past LEOP are flight telecommands.** Nothing on board advances the phase beyond 4 (LEOP): the team declares COMMISSIONING (5) and NOMINAL (6) with `SYS_SET_MISSION_PHASE(phase=N)` (S8.1 func 120; the FORCE flag permits a backwards move). The instructor console's `set_phase` (phases 0–6) is an instructor override for setting up or repairing a session, and `SET_PHASE` is not an alias of anything — expect the team to send func 120 at the end of LEOP-003 and COM-013.
- Instructor commands (`set_speed`, `freeze`, `resume`, `inject`, `clear_failure`, `override_passes`, `set_phase`, the `scenario_*` calls, breakpoint save/load) go over the instructor HTTP channel. That channel **rejects spacecraft telecommands with HTTP 403** — real TCs must go through the MCS.

5.10.1 Breakpoints (4)

Breakpoints are named full-state snapshots, saved and reloaded from the instructor UI (`/api/breakpoint/save`, `/api/breakpoint/load`). A save that cannot serialise every component fails outright rather than writing a partial snapshot.

ID	Name	Phase	Use
BP-1	Nominal Post-Commissioning	6 (NOMINAL)	Routine pass training (`scenarios/nominal_ops.yaml`)
BP-2	LEOP Day Start	3 (BOOTLOADER)	Day 3 (`scenarios/first_contact.yaml`)
BP-3	Commissioning Day Start	4 (LEOP complete)	Day 4 (`scenarios/commissioning_day_start.yaml` — instructor-authored, see below; not in the shipped set)
BP-4	What-If Branch	(any)	Day 5 replay — save a breakpoint before the decision point, then reload it from the instructor UI

BP-3 scenario YAML to add to the simulator

Written against the schema the loader actually accepts. `initial_conditions` supports only `spacecraft_phase`, `speed`, `override_passes`, `soc_pct`, `power_lines` (by line name), `params` (by numeric parameter id) and `failures`. Events must use `time_offset_s` (or a `condition`) plus one of the supported actions — `inject`, `clear`, `set_parameter`, `message`, `set_speed`, `set_phase`. Anything else is dropped at load time and raised as a `SCENARIO_CONFIG_INVALID` event, so a scenario that looks fine can silently do nothing. `difficulty` must be one of BASIC / BEGINNER / INTERMEDIATE / ADVANCED.

```
name: "Commissioning Day Start"
difficulty: INTERMEDIATE
duration_s: 14400          # 4 hours of compressed time across multiple passes
briefing: |
  EOSAT-1 has just exited LEOP. The S-band antenna is deployed, the
  application image is running, detumble has completed, time is set,
  and the spacecraft is in phase 4 (LEOP) holding COARSE_SUN.
  Your shift covers the first 4 hours of commissioning across
  ~4 ground passes (compressed).
  Goals for the shift:
    1. Complete the commissioning entry criteria and call for the
       phase advance to 5 (COMMISSIONING). The instructor performs
       that advance on the console — there is no telecommand.
    2. Run COM-001 through COM-008 in order.
    3. Power on the payload, cool the FPA to its -15 C setpoint, and
       complete COM-012 (First Light) before LOS at end of pass 4.

initial_conditions:
  spacecraft_phase: 4
  override_passes: true
  soc_pct: 90.0
  power_lines:
    ttc_tx: true
    htr_bat: true
    htr_obc: true
    aocs_wheels: true
    payload: false
    fpa_cooler: false
    prop: false
  params:
    0x020F: 3        # aocs.mode = COARSE_SUN
    0x0600: 0        # payload.mode = OFF

events:
  - time_offset_s: 600
    action: message
    params:
      text: "Optional: inject eps overcurrent on line 3 during COM-009 payload power-on"
  - time_offset_s: 5400
    action: message
    params:
      text: "Optional: inject tcs cooler_failure during COM-010 FPA cool-down"
```

```
expected_responses:
  - { category: command, description: "Report commissioning entry criteria met and request
  - { category: command, description: "Run COM-001 EPS Checkout" }
  - { category: command, description: "Run COM-005 AOCS Mode Transitions (COARSE_SUN 3 ->
  - { category: command, description: "Run COM-006 TTC Link Verification (LOW 16 kbps / HI
  - { category: command, description: "Run COM-008 FDIR Configuration — verify and exercis
  - { category: command, description: "Run COM-009 Payload Power On (EPS_POWER_ON line_ind
  - { category: command, description: "Run COM-010 FPA Cooler Activation (FPA_COOLER on=1,
  - { category: command, description: "Run COM-012 First Light — capture and assess one im
```

Optional anomalies are left as `message` cues rather than pre-armed injections so the instructor keeps the decision. Inject them live from the instructor UI when the moment arrives.

5.10.2 Anomaly scenario catalogue (36)

Every file below exists in `configs/eosat1/scenarios/`. Sim durations are the `duration_s` in the file.

File	Difficulty	Sim duration	What it does	Can pro
`aocs_actuator_stuck.yaml`	Medium	continuous	Seizes one reaction wheel (`rw_seizure`) — 4 wheels are flown, so this leaves 3	CTG CTG
`aocs_sensor_cascade.yaml`	Hard	2700 s	ST1 fails, +X CSS head fails, the magnetometer fails, then at T+600 s a **persistent 0.05 °/s gyro bias** on all three channels	CTG CTG
`aocs_star_tracker_failure.yaml`	Medium	continuous	Star tracker fails outright	CTG
`aocs_wheel_failure.yaml`	Hard	continuous	Wheel bearing failure mid-pass	CTG
`eps_anomaly.yaml`	Medium	varies	Generic EPS fault, often paired with `eps_undervoltage`	CTG CTG
`eps_overcurrent.yaml`	Easy	continuous	Sustained overcurrent on selected line	CTG
`eps_progressive_load_shed.yaml`	Hard	1800 s	Battery cell degrades, forces progressive shed	CTG
`eps_solar_panel_loss.yaml`	Hard	continuous	One solar array string fails	CTG
`eps_undervoltage.yaml`	Medium	continuous	Voltage drops below caution band	CTG
`first_contact.yaml`	(BP-2)	1800 s	LEOP Day Start scenario	—
`fpa_overtemp.yaml`	Medium	1200 s	FPA cooler under-spec: it cannot hold the commanded setpoint (default **−15 °C**; envelope −20 to 0 °C), so	CTG

File	Difficulty	Sim duration	What it does	Can pro
			`payload.cooler_margin_k` (0x064B) goes ≤ 0	
`gs_antenna_failure.yaml`	Very hard	continuous	Primary GS antenna fails mid-pass	CTG (**adv** — g only spa
`gs_rf_degradation.yaml`	Easy	600 s	BER climbs over the pass	CTG CTG
`gs_tracking_loss.yaml`	Medium	300 s	GS antenna loses tracking briefly	CTG
`nominal_ops.yaml`	(BP-1)	continuous	Healthy spacecraft for tools practice	—
`obc_bus_failure.yaml`	Hard	continuous	CAN bus A drops	CTG
`obc_crash.yaml`	Hard	one-shot	OBC application crash, reboot required	CTG
`obc_watchdog.yaml`	Hard	one-shot	Watchdog event mid-procedure	CTG
`obdh_bootloader_recovery.yaml`	Medium	continuous	Bootloader will not boot app SW	boo spe
`obdh_bus_isolation.yaml`	Medium	continuous	Subsystem CAN node isolation	CTG
`payload_corrupt_image.yaml`	Medium	one-shot	Next image returns corrupt	CTG
`payload_memory_failure.yaml`	Hard	continuous	Mass-memory ECC errors climbing	pay
`rw_bearing.yaml`	Medium	gradual	Wheel bearing wear, current climbs over hours	CTG
`separation.yaml`	(LEOP)	one-shot	Separation event	—
`separation_sequence.yaml`	(LEOP)	one-shot	Full separation through detumble	—
`sequential_checkout.yaml`	(commissioning)	continuous	LEOP-007 walk-through	—
`transponder_failure.yaml`	Hard	continuous	S-band transponder fails	CTG **EM** of Con
`ttc_no_tm_at_aos.yaml`	Easy	continuous	UHF beacon present but no S-band HK — the two are separate radios	CTG
`ttc_pa_overheat.yaml`	Medium	gradual	S-band power amplifier overheating	CTG
`debris_impact_hermes3.yaml`	Hard	one-shot	Conjunction / debris-impact case study	CTG NO
`scenario_2_cerise_adcs_fault.yaml`	Hard	continuous	Cerise-style ADCS fault case study	CTG

File	Difficulty	Sim duration	What it does	Can pro
`prop_slow_leak.yaml`	Medium	gradual	Propellant slow leak, tank pressure decaying	CTG
`prop_underperformance.yaml`	Medium	one-shot	Burn delivers less Δv than planned	CTG NOM repl
`prop_valve_stuck_open.yaml`	Hard	continuous	Isolation valve stuck open	CTG
`prop_burn_eclipse_power.yaml`	Hard	one-shot	Manoeuvre attempted into an eclipse power hole	CTG CTG
`prop_attitude_loss_mid_burn.yaml`	Very hard	one-shot	Attitude error exceeds the burn limit mid-firing; on-board abort	CTG CTG

EOSAT-1 flies propulsion. A 1U green monoprop, 66.9 m/s total, for debris avoidance, limited orbit maintenance and the 34.3 m/s end-of-life perigee-lowering burn (NOM-014: perigee 513 → 390 km, re-entry in 1.5 years at mean solar activity, 4.2 at solar minimum; natural decay alone would take 5.0 and 14.2 years). Propulsion telecommands are S8 funcs 108–118, telemetry base 0x0B00, HK SID **9**. A burn re-fits the SGP4 element set, so passes and eclipses genuinely move afterwards (1 m/s gives Δa +1.800 km and −33.5 s at +24 h), and time acceleration is clamped to ≤ 10× while a burn is firing. Do not describe this mission as having no propulsion.

Failure injections without a scenario file

The instructor UI can inject any failure the models implement directly, without a scenario. Ones worth knowing for this week:

Subsystem	Injection	What it does
eps	`sada_stuck`	Seizes the solar-array drive. `eps.solar_array_drive_angle` (0x0133) goes flat while the wing currents (0x0146 / 0x0147) start cycling with the orbit — generation drops to the fixed-wing duty
eps	`solar_panel_loss`, `bus_short`, `overcurrent`, `undervoltage`, `bat_cell`	the EPS ladder
aocs	`gyro_bias`	**Persistent** rate offset — see 18.10
aocs	`rw_seizure`, `rw_bearing`, `st_failure`, `st_blind`, `mag_failure`, `css_head_fail`, `mtq_failure`	the AOCS set
tcs	`cooler_failure`, `heater_stuck_on`, `heater_open_circuit`, `sensor_drift`	the thermal set
ttc		the link set

Subsystem	Injection	What it does
	`primary_failure`, `pa_overheat`, `high_ber`, `antenna_deploy_failed`, `gs_*` (ground-side)	
obdh	`watchdog_reset`, `obc_crash`, `bus_failure`, `stuck_in_bootloader`	the platform set

5.10.3 Multi-anomaly composition rules

If you chain two scenarios, the second triggers only after the first is acknowledged (or after a fixed delay if not). Recommended chains:

- `aocs_wheel_failure.yaml` + `transponder_failure.yaml` — tests multi-system anomaly response. Day 4, very hard.
- `eps_overcurrent.yaml` (payload line) + `payload_corrupt_image.yaml` — tests sustained anomaly + cascaded effect. Day 4, medium-hard.
- `gs_antenna_failure.yaml` (Iqaluit) + `gs_rf_degradation.yaml` (Troll next pass) — forces the team to operate with degraded ground assets only. Day 4, very hard.

Do not chain *three* anomalies unless the cohort is exceptional; the cognitive load exceeds control room training week's pedagogical intent.

Chapter 2 — Mission Simulation Day (Instructor Walkthrough)

> **Expected reading before this session.** Open textbook *Concurrent Spacecraft Design* (OER), ch. 19, §§19.1–19.6 (why agencies rehearse, structuring a rehearsal, role allocation, scripting the phases, the anomaly library, the debrief); this book, ch. 5.10 (scenario catalogue); course textbook, Part 2 (*CubeSat Operations*), §8.1 (pass cycle overview); ECSS-E-ST-70C *Ground systems and operations* (31 July 2008), §5.5 *Operations teams build-up and training* and §5.6 *Operational validation* (simulations and rehearsals with the operational simulator, §5.6.2 b).

Prerequisites: Sessions 5.1-5.2 (ground segment designed, operations concepts defined) **References:** ECSS-E-ST-70-11C (Operability), ECSS-E-ST-70-32C (Procedures), ESA OPS-G Training Manual, NASA Mission Operations Directorate Training Handbook

Learning Objectives

By the end of this session, participants will be able to: 1. Execute a simulated LEOP sequence from separation through first telemetry 2. Perform commissioning activities using operational procedures 3. Respond to injected anomalies using the FDIR architecture and contingency procedures 4. Document anomaly response in real-time using structured forms 5. Conduct nominal science operations with live data budget tracking

1. Simulation Overview & Roles

This session is a **hands-on simulation** of mission operations. The facilitator acts as the "Universe" -- injecting events, anomalies, and time jumps. The team operates the mission using SpaceCDF and their procedures from Chapter 5.2.

Role Assignments

Role	Responsibility	SpaceCDF Feature Used
Flight Director (FD)	Overall decision authority; coordinates team	Dashboard, all tabs
Spacecraft Controller (SC)	Sends commands; monitors housekeeping telemetry	ConOps Editor, mode control
Ground Station Operator (GSO)	Manages antenna scheduling; link acquisition	Data Budget, Link Budget
Payload Operator (PO)	Plans and executes science observations	Payload parameters, data budget
Flight Dynamics (FD-nav)	Monitors orbit; plans and calls the manoeuvres (EOSAT-1 has a 1U green monoprop, 66.9 m/s)	Orbit parameters, PROP tab, sustainability
Anomaly Logger	Documents all events, anomalies, and decisions	Worksheet 5.3

For smaller teams, combine roles: SC+GSO and PO+FD-nav are natural pairings.

Simulation Timeline

Sim Time	Real Time	Phase	Events
T+0 to T+1 hr	0-20 min	LEOP	Separation, beacon acquisition, deploy, health check
T+1 hr to T+24 hr	20-35 min	Early LEOP	SA deploy, ADCS init, comms chain validation
T+1 day to T+7 days	35-50 min	Commissioning	Subsystem checkout, first payload activation
T+7 days to T+30 days	50-65 min	Early Operations	Nominal science, orbit maintenance
Anomaly injection	65-85 min	Contingency	Anomaly response exercise
T+30 days to T+1 year	85-100 min	Nominal Operations	Routine operations, data review
Debrief	100-120 min	Wrap-up	Lessons learned, documentation review

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2. LEOP Simulation

The facilitator narrates the simulation. Time is compressed. Each team executes their LEOP procedure.

Facilitator Script -- LEOP

[T+0 -- Separation] *"Your satellite has separated from the deployer. Deployment switches have released. The 30-minute timer has started. You are in radio silence -- no RF emissions allowed until the timer expires. What is your spacecraft doing right now?"*

Worked response: Timer counting down. Battery providing power. No subsystems active except OBC running timer.

[T+30 min -- Timer Expires] *"The 30-minute timer has expired. Your OBC has commanded antenna deployment. What do you expect to see?"*

Expected: Antenna deployment command sent. Beacon should begin transmitting.

[T+35 min -- First Pass Over Ground Station] *"Your ground station at [location] has line of sight. The antenna operator reports: carrier detected at [frequency] with Doppler of -25 kHz. Signal strength is -110 dBm. Is this what you expected?"*

Teams should verify: - Carrier frequency matches expected (accounting for Doppler) - Signal strength is consistent with link budget at current range - Beacon data rate is as designed

[T+40 min -- First Telemetry] *"Beacon lock achieved. You are receiving housekeeping telemetry. I am going to read you the first HK frame:"*

Parameter	Mnemonic (id)	Value	Expected Range	Status
Bus voltage	`eps.bus_voltage` (0x0105)	32.4 V	29.6 - 34.6 V	Nominal
Battery SoC	`eps.bat_soc` (0x0101)	88 %	> 50 %	Nominal

Parameter	Mnemonic (id)	Value	Expected Range	Status
Power generation	`eps.power_gen` (0x0107)	12 W	low while tumbling	Nominal
OBC temperature	`tcs.temp_obc` (0x0406)	22 C	-10 to +50 C	Nominal
Battery temperature	`tcs.temp_battery` (0x0407)	19 C	+5 to +35 C	Nominal
AOCS mode	`aocs.mode` (0x020F)	2 (DETUMBLE)	Expected	Nominal
Body rates	`aocs.rate_roll/pitch/yaw` (0x0204-06)	1.4 deg/s	Decreasing	Nominal
Running image	`obdh.sw_image` (0x0311)	0 (BOOTLOADER)	Expected pre-boot	Nominal
Antenna status	`ttc.antenna_deployed` (0x0520)	0 (stowed)	deployed in COM-006	--
Beacon	`ttc.beacon_active` (0x0527)	1	Expected	Nominal
Link status	`ttc.link_status` (0x0501)	2 (LOCKED)	2 = LOCKED	Nominal
Uplink status	--	Not yet attempted	--	--

Note the bus: EOSAT-1 runs an **8S, 100 Wh** pack (GomSpace BPX) — 24.0 V empty, 28.8 V nominal, 33.6 V full — on an unregulated bus that follows the battery up to a 33.4 V cap, so 88 % SoC reads 24.0 + 9.6 × 0.88 ≈ 32.4 V. A 7-8 V reading is a 2S CubeSat, not this spacecraft. The under-voltage flag and the in-model load shed fire at **28.4 V** and FDIR EPS safe mode at **27.7 V** (`simulation.yaml` `eps.bus_protection`).

"All parameters nominal. Proceed with LEOP sequence."

[T+1 hr -- Uplink Validation] *"You now attempt your first uplink command. Send a 'NOP' (no operation) command to verify the uplink chain."*

Teams should: compose command, verify encoding, transmit, wait for acknowledgement in telemetry.

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3. Commissioning Simulation

Facilitator Script -- Commissioning

[Time Jump: T+24 hr] *"Your satellite has completed detumbling. AOCS is in COARSE_SUN (mode 3). All HK parameters are nominal after 8 ground passes. You are ready to begin commissioning. What is your first activity?"*

Expected: EPS and thermal checkouts (COM-001, COM-002), then the AOCS actuator checkout (COM-004): enable the four wheels and command wheel-controlled NOMINAL. Only then the AOCS sensor calibration (COM-003): gyros and a star tracker into the loop, then the **on-board** gyro bias calibration with `AOCS_GYRO_CALIBRATION` (S8 func 13). It is refused above 0.10 °/s of body rate, which the wheels-off COARSE_SUN hold cannot meet — that is why COM-004 comes first (ruling Q16). There is no ground-computed bias upload, and no magnetometer calibration (single unit, nothing to select).

[T+3 days -- Payload First Light] *"AOCS is calibrated and holding NOMINAL (mode 4) at 0.3 degree attitude error. You are ready for payload first light. Walk me through your procedure."*

Teams execute their science observation procedure from Worksheet 5.2.

[T+5 days -- First Data Download] *"Payload has captured 150 MB of imagery. Your next pass is 12 minutes long over [station]. Can you download all the data in one pass?"*

Teams should calculate: - Data rate x pass duration x protocol efficiency = available capacity - Compare to 150 MB -- determine if multiple passes needed - Plan the download sequence

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4. Anomaly Injection Scenarios

The facilitator selects 2-3 anomalies from the list below, appropriate to the mission design. Teams respond using their FDIR rules and contingency procedures.

Anomaly Menu (Facilitator selects 2-3)

Anomaly A: Battery Under-Voltage

"During eclipse, you receive an alert: bus voltage has dropped to 27.4 V, below the 27.7 V FDIR EPS safe-mode threshold (the caution band, the under-voltage flag and the in-model load shed all start at 28.4 V). Battery temperature is 5 C. SoC is estimated at 18%."

Expected response: 1. Verify FDIR has triggered load shedding (payload off, non-essential heaters off) 2. Check power budget: is consumption > generation? 3. Possible causes: SA degradation, unexpected load, battery degradation, long eclipse 4. Immediate action: Verify safe mode; disable non-essential loads manually 5. Investigation: Review power trends over last 24 hours; compare to power budget

Anomaly B: Communication Loss

"You have not received telemetry for the last 3 scheduled passes (18 hours). The ground station reports pointing and frequency are correct but no carrier detected."

Expected response: 1. Verify ground station equipment (antenna, LNA, modem) -- rule out ground fault 2. Check with other ground stations (SatNOGS) for any signal detection 3. Consider causes: TX failure, antenna deployment failure, attitude loss (antenna not pointing) 4. Check whether the UHF beacon is still being heard. It is a separate radio on the non-switchable `ttc_rx` line: **PA-off kills the S-band TM while the beacon survives.** Beacon present + S-band silent isolates the fault to the S-band chain. 5. On-board autonomy: the 24-hour no-contact timer will switch to the redundant transponder (FDIR TTC-01) 6. Plan: blind-command the TX chain per CTG-019; if still nothing after two consecutive passes, escalate EMG-004 (Loss of Communication)

Anomaly C: ADCS Anomaly -- Loss of Fine Pointing

"`aocs.att_error` (0x0217) has increased from 0.1 deg to 5 deg. The selected star tracker reports status 4 (failed). `aocs.momentum_saturation_pct` (0x0282) reads 80% -- near saturation."

Expected response: 1. Diagnose: which head? `aocs.st1_status` (0x0240) / `aocs.st2_status` (0x0243), codes 0 off / 1 boot / 2 tracking / 3 blind / 4 failed. Optical contamination? Sun in FOV? 2. There are **two** trackers -- bring up the other one (`ST2_POWER` func 5 / `ST1_POWER` func 4) and `ST_SELECT` (func 6) before assuming attitude is lost. 3. Check the estimator source bitmask `aocs.est_source` (0x02C1): bit0 ST, bit1 CSS, bit2 MAG, bit3 gyro. 4. Command desaturation with `AOCS_DESATURATE` (S8 func 1). Budget **20-50 min** for a full dump, not five minutes; imaging is inhibited for the duration and the spacecraft returns to its pre-desat mode. Watch `aocs.h_body_mag` (0x02B9) and

`aocs.desat_ineffective` (0x029C). 5. If both heads are gone: `AOCS_SET_MODE(mode=3)` COARSE_SUN, the sun-safe hold. 6. Impact assessment: can science continue on CSS + magnetometer + gyro only?

Anomaly D: Unexpected Safe Mode Entry

"Your satellite has autonomously entered Safe Mode. Last telemetry before entry showed: OBC reboot counter incremented by 3 in 10 minutes. All other parameters were nominal."

Expected response: 1. Investigate: Multiple reboots suggest SEU (Single Event Upset) or firmware bug 2. Check radiation environment: Was the satellite in SAA (South Atlantic Anomaly)? 3. Review FDIR logs (stored in non-volatile memory) for fault triggers 4. Recovery plan: `OBC_SET_MODE(mode=0)` (S8 func 50) back to NOMINAL and `AOCS_SET_MODE` back to the operational mode; re-issue `HK_ENABLE` for any SID lost to a reboot; monitor for recurrence. Note the FDIR rule: `obdh.reboot_count` > 4 escalates to spacecraft emergency. 5. If recurring: Upload software patch to increase SEU robustness

Anomaly E: Thermal Exceedance

"`payload.fpa_temp` (0x0601) has climbed to +19 C during a science observation, outside the FPA health band of −25.5 to +17.5 C, and `payload.fpa_ready` (0x0610) has dropped to 0. `tcs.temp_battery` (0x0407) is 43 C — past the 42 C FDIR heater-off trip and approaching the 45 C over-temperature event."

Expected response: 1. Immediate: Disable payload to stop heat generation 2. Check attitude: Is the hot radiator face pointing toward the sun? 3. Review thermal model: Was this orbit geometry (high beta angle) predicted? 4. Mitigation: Adjust duty cycle; plan observations for cooler orbit phases 5. If structural: May need to update thermal model and operational constraints

Anomaly Documentation

For each injected anomaly, teams complete the **Anomaly Response Form** on Worksheet 5.3:

Field	What to Record
Time (sim)	When the anomaly was detected
Observation	Exact telemetry values and symptoms
Initial assessment	**FDIR severity 0–4** (how serious the event is). Record separately from the **autonomy level 0–3** (who acted: 0 hardware, 1 subsystem SW, 2 OBC FDIR, 3 ground). The two scales are different things and must never be conflated.
Diagnosis	Suspected root cause
Action taken	Commands sent, procedures executed
Result	System state after response
Follow-up required	Additional investigation, procedure updates

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5. Nominal Operations & Debrief

Nominal Operations

[Time Jump: T+30 days to T+1 year] *"Your mission has been operational for one month. All commissioning activities are complete. You are now in nominal operations. Let's review your operational metrics."*

Teams report: - Total data downlinked in 30 days vs plan - Number of science observations completed vs plan - Any anomalies encountered (from injection exercise) - Current budget status: power margin (nominal orbit-average load ≈ 22.7 W against ≈ 45.1 W generation at the worst-generation beta, β = −31.2°), data margin, propellant remaining out of the 66.9 m/s total - Orbit status: any conjunction alerts? Debris compliance on track?

Debrief

1. **What went well?** - Which procedures worked as written? - Which FDIR rules triggered correctly? - Where did the team communicate effectively?
2. **What could be improved?** - Which procedures needed real-time modification? - Were there gaps in the FDIR design? - Where did decision-making slow down?
3. **Lessons for the design:** - Did the simulation reveal any design weaknesses? - Should any requirements be changed based on operational experience? - What additional autonomy would help?

Facilitator Notes

Preparation

- Review each team's ConOps, FDIR rules, and procedures from Chapter 5.2
- Select anomalies appropriate to the point the team has reached (e.g. don't inject a payload anomaly before COM-009 has powered the payload). EOSAT-1 *does* have propulsion, so the `prop_*` scenarios are all in play once NOM-004 has been briefed.
- Prepare HK data tables for each phase (modify the values in Section 2 to match the team's mission parameters)
- Have backup anomalies ready if teams resolve injections quickly

Pacing

- LEOP simulation should feel time-pressured (short passes, decisions needed quickly)
- Commissioning can be more relaxed
- Anomaly injection should be challenging but solvable with the team's procedures
- Allow extra time for debrief -- this is where the deepest learning occurs

Assessment Criteria

Observe and note (for feedback, not formal grading): - Did the team follow their procedures or improvise? - Did the Flight Director coordinate effectively? - Was the anomaly response systematic (detect-diagnose-plan-execute-verify)? - Did the team document events in real-time? - Did anyone identify a design flaw during operations?

Session Summary

Topic	Key Takeaway
LEOP	First 72 hours are critical; every step is time-constrained

Topic	Key Takeaway
Commissioning	Systematic subsystem checkout before payload operations
Anomaly response	Follow the process: detect, assess, diagnose, plan, execute, verify, document
Documentation	Real-time logging is essential -- memory is unreliable under stress
FDIR validation	Simulation reveals gaps in FDIR design that analysis alone cannot find
Team coordination	Clear roles, communication protocols, and decision authority are essential
Design feedback	Operational experience should feed back into design (requirements, FDIR, procedures)

Anomaly injection menus (moved from textbook Chapters 10 and 11)

These sections are instructor-only: they list the scenario YAMLs, cue moments, and coaching protocols for anomaly injection during simulation days. Trainees on console should not see them in advance.

LEOP anomaly injection menu

10.6 Anomaly injection menu (instructor)

LEOP shift coaching

10.8 Coaching during the shift

Commissioning anomaly injection menu

11.6 Anomaly injection menu

Chapter 18 — Anomaly Cue Cards

For each scenario in the Day 3 and Day 4 instructor menus, this chapter provides a one-page detailed cue card the instructor uses during the sim and the team uses during debrief.

Card format:

- **Scenario file**
- **Cue moment** (what in the pass should be happening when injection fires)
- **First indicator** (what the operators will see first)
- **Recovery procedure** (the canonical response)
- **Pass criterion** (what defines a successful recovery)
- **Five debrief questions** to ask the team after the pass

18.1 No TM at Pass Start (Easy)

- **Cue moment.** Inject at scenario T+0, before any operator command. The spacecraft is in BOOTLOADER_OPS; the beacon is present but the OBC will not produce full HK even after OBC_BOOT_APP.
- **First indicator.** SPACON sees `obdh.sw_image` (0x0311) correctly flip to 1 after step 5, but no SID 1 packets arrive in the 1-second interval that follows, and the SIDs the operator then commands on with `HK_ENABLE` (**S3.5**) — 3 (TCS, 60 s), 4 (Platform, 8 s), 6 (TT&C, 8 s) — never appear either. Note the trap that is *not* the anomaly: application boot re-applies `default_enabled_sids: [1, 11]`, so only EPS and the Beacon are on by default. A team that has simply forgotten to issue `HK_ENABLE` has not found a fault.
- **Recovery procedure.** CTG-019 No Telemetry at Pass Start.
- **Pass criterion.** Team correctly identifies that the spacecraft is responding to commands (S1 acks present) but TM flow is broken; they invoke CTG-019 within 60 seconds of the missing TM; they continue the procedure with blind commanding (using S1 acks only as confirmation).
- **Debrief Qs.** 1. How did SPACON realise full HK wasn't arriving (other than waiting for it)? 2. Did the team correctly distinguish *no TM* from *no commanding*? 3. Did anyone propose invoking EMG-004 (Loss of Communication)? Was that justified? (EMG-003 is OBC Reboot, a different card.) 4. At what time did the team decide to continue with blind commanding rather than wait? 5. What was the impact on subsequent procedure steps?

18.2 Power-Line Overcurrent (Easy — Day 3 or 4)

- **Cue moment.** During LEOP-007 sequential power-on, inject on the line that's about to be enabled — `aocs_wheels` (`line_index=7`, step 3) — or during COM-009 payload power-on, on `payload` (`line_index=3`). Line indices run 0–8: 0 obc, 1 ttc_rx, 2 ttc_tx, 3 payload, 4 fpa_cooler, 5 htr_bat, 6 htr_obc, 7 aocs_wheels, 8 prop.
- **First indicator.** EPS sees the per-line current (`eps.line_current_N`, 0x0118+N) go over its trip threshold (payload 2.5 A, aocs_wheels 0.8 A) for the 50 ms trip time; `eps.oc_trip_flags` (0x010D) sets the bit and the line is opened autonomously.
- **Recovery procedure.** CTG-012 Overcurrent Response.
- **Pass criterion.** Team acknowledges the FDIR rule within 20 s; identifies the affected line; files anomaly ticket; does NOT re-enable the line in the same pass (the autonomous action is correct — leave it off; investigate next shift).

- **Debrief Qs.** 1. Did the team recognise that FDIR had already opened the line, or did they try to issue `EPS_POWER_OFF(line_index=N)` (S8 func 20) manually? Did anyone remember that re-enabling needs `EPS_RESET_OC_FLAG(line_index=N)` (func 21) *first*? 2. Did anyone propose re-enabling the line in the same pass? 3. Was the anomaly ticket filed before pass end? 4. What downstream procedure steps were affected? 5. Should the team have invoked the autonomous response manually instead of letting FDIR fire (a defensibility question — both answers are right depending on context)?

18.3 Reaction Wheel Stuck (Medium)

- **Cue moment.** During COM-004, once the wheels are enabled and the spacecraft is in wheel-controlled NOMINAL (the wheels stay disabled through LEOP). Inject `rw_seizure` on one wheel as it spins up.
- **First indicator.** AOCS sees that wheel's speed (`aocs.rw2_speed` , 0x0208 for wheel index 1) pinned at 0 RPM while its current (`aocs.rw2_current` , 0x0251) stays high — the motor trying to turn a stuck rotor — and its bearing health (`aocs.rw2_bearing_health` , 0x0285) collapsing.
- **Recovery procedure.** CTG-026 AOCS Actuator Stuck Recovery / CTG-007 Reaction Wheel Anomaly.
- **Pass criterion.** Team identifies the affected wheel within 60 s; disables it with `AOCS_DISABLE_WHEEL(wheel_idx=N)` (S8 func 2); confirms `aocs.wheels_enabled` (0x0260) = 3; verifies detumble continues (B-dot runs on the **magnetorquers**, not the wheels, so a stuck wheel does not stop it); files anomaly ticket.
- **Debrief Qs.** 1. Did AOCS catch the stuck wheel from current draw, or only from the FDIR rule? 2. Was the FD's GO to disable the wheel polled correctly? 3. Did the team correctly choose to continue detumble rather than abandoning — and did anyone spot that detumble never needed the wheels in the first place? 4. Could the team predict the impact on the pointing budget with 1 of 4 wheels down? (Very little in the mean: a 3-of-4 pyramid still spans R^3 and FINE_POINT holds 0.0188° against 0.0208° on four. What is lost is the null space — bias-speed regulation and zero-crossing avoidance — so short-term jitter is worse, and the mode is flagged at `aocs.fine_degraded` 0x0299 with the deadband relaxed 4×.) 5. What's the long-term implication for the mission (e.g. reduced agility, more frequent desat)?

18.4 Star Tracker Failure (Medium — Day 3 or 4)

- **Cue moment.** Late in LEOP-005 sun acquisition (Day 3), or during the COM-005 NOMINAL transition (Day 4).
- **First indicator.** AOCS sees `aocs.st1_status` (0x0240) go to **3 (blind)** or **4 (failed)** — the codes are 0 off, 1 boot, 2 tracking, 3 blind, 4 failed — and `aocs.att_error` (0x0217) starts climbing. `aocs.est_innov_rej_count` (0x02C2) rises as the estimator gates the bad fixes.
- **Recovery procedure.** CTG-008 Star Tracker Failure.
- **Pass criterion.** Team identifies the failure within 30 s; **brings up the second head** (`ST2_POWER(on=1)` , S8 func 5) and selects it (`ST_SELECT` , func 6) before treating attitude as lost; powers the failed head off (`ST1_POWER(on=0)` , func 4); re-initialises the estimator with `AOCS_EST_RESET(mode=1)` (func 91, warm) if the solution has diverged; only if *both* heads are gone commands COARSE_SUN (`AOCS_SET_MODE(mode=3)`); files anomaly ticket.
- **Debrief Qs.** 1. Did AOCS distinguish blind (3) from failed (4), and what's the operational difference? 2. Did anyone reach for the redundant head first, or did the team jump straight to a safe-mode transition? This is a two-tracker spacecraft. 3. Did the team consider deferring imaging if ST2 also fails? 4. What's the pointing budget on CSS + magnetometer + gyro only? (Degrades to roughly a degree — the COARSE_SUN class of performance, not the sub-0.1° FINE_POINT budget.) 5. Mission

impact: how many days can the spacecraft operate in degraded star-tracker mode before science deliverables are missed?

18.5 OBC Watchdog Reset (Hard)

- **Cue moment.** Mid-LEOP-007, around steps 3–5 (AOCS suite line, magnetometer and magnetorquers, DETUMBLE). Inject `watchdog_reset` on obdh.
- **First indicator.** All telemetry stops for ~15 s; then the UHF beacon comes back with the OBC in BOOTLOADER (`obdh.sw_image` 0x0311 = 0), `obdh.reboot_count` (0x030A) incremented and `obdh.last_reboot_cause` (0x0316) = 1 (watchdog).
- **Recovery procedure.** CTG-010 OBDH Watchdog Recovery.
- **Pass criterion.** Team recognises the unexpected reboot within 30 s of TM resumption; invokes CTG-010; re-establishes application boot (`OBC_BOOT_APP` , S8 func 55 — LEOP-001 step 5); **re-issues `HK_ENABLE` (S3.5) for every SID except 1 and 11**, because application boot re-applies `default_enabled_sids: [1, 11]` ; restores time sync (`SET_TIME` S9.1); re-clears beacon mode and re-selects the LOW tier; logs every step lost; files anomaly ticket.
- **Debrief Qs.** 1. Did SPACON immediately recognise the bootloader beacon pattern? 2. Did the team avoid panic-commanding while the OBC was mid-reboot? 3. How long from reboot to spacecraft fully recovered? (Industry: < 5 min for a clean watchdog recovery.) 4. Did the team correctly identify which procedure steps needed re-execution vs which were preserved? 5. What's the right way to *prevent* this in future shifts? (Hint: ensure watchdog kick is in the procedure pre-conditions for long-running activities.)

18.6 OBC Bus Failure (Hard)

- **Cue moment.** During LEOP-007, after step 3 (AOCS suite line energised, AOCS HK SID 2 appearing).
- **First indicator.** AOCS HK SID 2 stops flowing; OBDH HK (SID 4) shows `obdh.bus_a_status` (0x030F) = 1 (ERROR) with `obdh.active_bus` (0x030E) = 0 (A).
- **Recovery procedure.** CTG-017 Bus Failure Switchover.
- **Pass criterion.** SPACON recognises bus A failure within 30 s; commands **`OBC_SELECT_BUS(bus=1)` (S8 func_id 54)** — *not* `OBC_SWITCH_UNIT` (func 53), which fails over the OBC unit and is CTG-011; verifies `obdh.active_bus` = 1, bus B healthy and AOCS HK resumed; files ticket.
- **Debrief Qs.** 1. How did SPACON distinguish *no AOCS HK* from *bus A failure*? 2. Was the FD GO for switchover authorised cleanly? 3. Was the recovery faster than the watchdog recovery? Why should it be? 4. Mission impact: are we now running on the redundant bus for the rest of the mission? (Yes — bus B can't fail back without ground intervention.) 5. What's the long-term consequence (no more bus redundancy)?

18.7 Ground Station Antenna Failure (Very Hard — Day 4)

- **Cue moment.** Mid-pass over Iqaluit, ideally during a commanded action.
- **First indicator.** Ground Seg sees Iqaluit antenna gain drop; baseband Eb/N0 collapses; TT&C sees BER climb and rx lock loss.
- **Recovery procedure.** CTG-020 Ground Station Antenna Failure — an **advisory** card. It contains no spacecraft telecommands: this is handled entirely on the ground side.
- **Pass criterion.** Team correctly identifies this as a ground-side failure (not spacecraft) — `ttc.gs_equipment_status` (0x0524) is the bitmask, and a healthy spacecraft is still radiating; recognises that the current pass is effectively lost; re-plans the remaining activities onto the next

station in the five-station network (Iqaluit, Troll, Svalbard, Inuvik, O'Higgins); files ticket; informs customer (Mission Manager).

- **Debrief Qs.** 1. Did the team distinguish ground-antenna failure from spacecraft TT&C failure? 2. Did anyone propose escalating to EMG-004 (Loss of Comm)? Was that justified? (It should not be — the spacecraft is fine.) 3. Was the re-plan onto the next station done on the ground schedule, or did someone try to command the spacecraft for a ground fault? 4. What was the impact on commissioning timeline? Did the team miss any time-critical activity (e.g. an eclipse window for FPA cool-down)? 5. Mission Manager: how was the customer bulletin worded?

18.8 FPA Cooler Under-Spec (Medium)

- **Cue moment.** During COM-010 FPA cooler activation. The loop normally closes in about **10 minutes**, so inject at T+3 to T+5 min, while the FPA is still on the ramp between +4 C and -8 C.
- **First indicator.** Thermal sees the FPA cool-down flatten short of the setpoint; `payload.fpa_ready` (0x0610) never sets; `payload.cooler_margin_k` (0x064B) goes **≤ 0** and `payload.cooler_floor_c` (0x064A) rises above the commanded setpoint. The mirror pair `tcs.fpa_cooler_floor_c` (0x0424) and `tcs.fpa_cooler_margin_k` (0x0425) tell the same story on the thermal tab.
- **Recovery procedure.** CTG-004 Thermal Exceedance + payload-safe.
- **Pass criterion.** Team accepts that the cooler is under-spec and understands *why the answer is a warmer setpoint, not more waiting* — the cooler is a miniature Stirling-class machine at ~23 % of Carnot, so its authority falls as the lift grows and a negative margin means the setpoint is physically unreachable at the present sink. Decides whether to (a) command a warmer setpoint with `PAYLOAD_COOLER_SETPOINT` (S8 func 36, uint16 in **centi-kelvin**, envelope -20 to 0 C) and continue with degraded image quality, or (b) abandon COM-010 and re-plan; files ticket; if continuing, runs COM-011 cal and COM-012 First Light with the warmer FPA.
- **Debrief Qs.** 1. Did Thermal recognise the flattening cool-down from the trend line, or only when `fpa_ready` failed to set? 2. Did the FD make a clear go/no-go decision on continuing? 3. Did anyone read `cooler_margin_k` and realise that waiting longer could never work? What SNR cost does the warmer setpoint carry (`payload.snr` , 0x0616)? 4. If continuing, was the customer bulletin updated to flag the science quality impact? 5. What's the long-term mission impact if the cooler is permanently under-spec?

18.9 Corrupt Image Acquired (Medium)

- **Cue moment.** During the COM-012 capture step (`PAYLOAD_CAPTURE` , S8 func 28), inject the corruption so the resulting image fails its quality assessment.
- **First indicator.** Payload sees S5 event **0x0607 IMAGE_CHECKSUM_ERROR** after the product is stored, and `payload.checksum_errors` (0x0609) increments.
- **Recovery procedure.** CTG-015 Corrupted Image Recovery.
- **Pass criterion.** Team marks the image as failed; re-queues for re-acquisition next pass; investigates root cause (typically FPA temperature, cosmic ray hit during integration, or pointing wobble); files ticket; if re-acquisition succeeds next pass, First Light declared.
- **Debrief Qs.** 1. Did Payload correctly recognise the corruption from the 0x0607 event, or did they only realise from the quality assessment? Did they delete before re-capturing (`PAYLOAD_DELETE_IMAGE` , func 30, then `PAYLOAD_CAPTURE` , func 28)? 2. Was the re-acquisition triggered cleanly (i.e. did the team re-plan rather than re-running the failed command blindly)? 3. What's the root-cause hypothesis? (Usually cosmic ray for a one-off; FPA T for a recurring

failure.) 4. Did the team update the customer bulletin to reflect the deferred First Light? 5. How many image attempts is the team willing to make in one shift before formally deferring First Light?

18.10 AOCS Sensor Cascade (Hard)

This is `aocs_sensor_cascade.yaml`, and it is where **"IMU drifts" is real**. The scenario runs 2700 s and fires four injections:

T+	Injection	Effect
60 s	`st_failure` unit 1	ST1 → status 4; on-board FDIR should fall to ST2, but the operator must verify
180 s	`css_head_fail` face px	+X coarse sun sensor head dies; composite sun vector degrades
300 s	`mag_failure`	the single magnetometer fails (no B unit, no switch exists); operator must recognise the loss of B-dot and magnetorquer desaturation and protect momentum
600 s	`gyro_bias` axis=all bias=0.05	**0.05 °/s bias on all three gyro channels**

The gyro bias is the teaching point and it needs describing precisely. It is written into the **persistent injected offset** (`aocs.gyro_bias_injected`, 0x0296 — instructor visibility), not into the Gauss-Markov truth state, so it does **not** mean-revert over the 1800 s bias time constant and it **survives ticks and breakpoint save/load until it is explicitly cleared**. If you reload a breakpoint expecting a clean spacecraft, clear the failure first.

0.05 °/s is about 1800× the 0.1 °/h bias instability the MEKF is built from — and it is **survivable while a star tracker is healthy**: the filter's bias state absorbs a constant offset. The operator's only symptoms are a step in the innovation-rejection counter (`aocs.est_innov_rej_count`, 0x02C2) and `aocs.gyro_bias_x/y/z` (0x0270–0x0272) diverging from the estimate. It becomes mission-ending the moment the tracker is also lost before the filter has re-converged, because ECLIPSE and every propagate-only phase then drift at 0.05 °/s = **3 °/min**.

Note that the gyros are three single-axis units with **no redundant unit**, there is no IMU power line and no IMU cold-boot delay, and gyro bias calibration is on-board (`AOCS_GYRO_CALIBRATION`, S8 func 13) — not a ground-computed upload.

- **Cue moment.** Start the scenario just before COM-011 (Payload Calibration), so the gyro injection at T+600 s lands around the FINE_POINT transition.
- **First indicator.** `aocs.st1_status` (0x0240) → 4 (failed); then the sun-vector and magnetometer degradations; then `aocs.est_innov_rej_count` (0x02C2) steps up and `aocs.att_error` (0x0217) climbs.
- **Recovery procedure.** CTG-008 Star Tracker Failure → CTG-002 AOCS Anomaly (± CTG-025 AOCS Sensor Loss Recovery) + defer First Light.
- **Pass criterion.** Team works the failures in order: verifies the fall-back to ST2 (`ST_SELECT`, func 6) rather than assuming it; recognises that the magnetometer has no spare (there is no `MAG_SELECT`) and protects momentum; reads `aocs.est_source` (0x02C1) to see what the estimator is actually using; identifies the gyro bias from 0x0270–0x0272 and the rejection counter; recovers gyro knowledge with `AOCS_EST_RESET(mode=1)` (S8 **func 91**, warm) once the bias is bounded, or

accepts ST-only determination. If the pointing budget is broken, commands COARSE_SUN (`AOCS_SET_MODE(mode=3)`), formally defers First Light, files a ticket, tells Mission Manager.

- **Debrief Qs.** 1. Did the team correctly chain the procedures (CTG-008 for the tracker, CTG-002 for the AOCS response)? 2. Did anyone notice the gyro bias *before* the star tracker work was finished — and did they understand why the healthy tracker was hiding it? 3. What happens to this spacecraft at the next eclipse entry with the bias uncleared? (Propagate-only at 3 °/min.) 4. Did anyone propose attempting First Light anyway? Was that justified or risky? 5. Was the formal-deferral decision made by the FD with clear team agreement, or unilateral?

18.11 Progressive Battery Load-Shed (Hard)

- **Cue moment.** During COM-010 (FPA cooler activation), early in the cool-down.
- **First indicator.** EPS sees `eps.bus_voltage` (0x0105) trending down unexpectedly and the SoC dropping at roughly twice the predicted rate. The ladder the team will walk down: under-voltage flag, bus warning event (0x0103) and in-model shed at **28.4 V** (order payload → fpa_cooler → ttc_tx → aocs_wheels → prop), bus alarm event (0x0104) and FDIR EPS safe mode at **27.7 V** — all from `simulation.yaml` `eps.bus_protection` . SoC-driven shed stages sit at 30 / 20 / 10 % (`eps.load_shed_stage` , 0x0134).
- **Recovery procedure.** CTG-001 Under-Voltage Load Shed (CTG-023 Progressive Load Shed for the staged version).
- **Pass criterion.** Team sheds progressively in priority order — payload OFF (`PAYLOAD_SET_MODE(mode=0)`), cooler OFF (`FPA_COOLER(on=0)` , func 43), PA off (`TTC_PA_OFF` , func 67), wheels line off (`EPS_POWER_OFF(line_index=7)`) — watching for recovery between each, rather than dumping everything at once; keeps the OBC and receive chain (lines 0 and 1, both non-switchable) alive; recognises that the auto-restore walks the lines back one per tick once the bus clears 28.9 V; files ticket; informs Mission Manager.
- **Debrief Qs.** 1. Did EPS catch the unexpected SoC trend before FDIR fired, or only after? 2. Was load-shed escalation correct (priority 1 first, watch, then escalate)? 3. Did the team weigh shedding the FPA cooler — which throws away the cool-down work — against the fact that it is the single biggest switchable load at 15 W? 4. Was the FD GO chain correct for each escalation? 5. Root-cause hypothesis? (Usually battery cell degradation in real missions.)

18.12 `aocs_wheel_failure.yaml` + `transponder_failure.yaml` (Very Hard — Day 4)

This is the multi-anomaly scenario — two faults in quick succession.

- **Cue moment.** Wheel failure injected mid-Pass 3 during imaging; transponder failure injected at start of Pass 4.
- **First indicator (wheel).** AOCS sees that wheel's current (`aocs.rwN_current` , 0x0250–0x0253) spike and its speed (0x0207–0x020A) drop to 0; `aocs.wheels_enabled` (0x0260) falls to 3.
- **First indicator (transponder).** TT&C sees the S-band link drop at Pass 4 AOS — `ttc.link_status` (0x0501) ≠ 2, no S1 acks, no TM — **while the UHF beacon keeps arriving** (HK SID 11, `ttc.beacon_active` 0x0527 = 1). That contrast is the tell: the beacon is a separate radio on a separate band.
- **Recovery procedures.** CTG-007 (wheel) → CTG-003 (link).
- **Pass criterion.** Team handles the wheel failure during Pass 3, makes a deliberate decision about Pass 4 (defer imaging? image on the remaining three wheels, accepting `fine_degraded` ?), then when the transponder fails at AOS, recognises the second anomaly, invokes CTG-003 link recovery, and re-establishes the link before LOS.

- **Debrief Qs.** 1. Did the team correctly handle the first anomaly without letting it cascade? 2. When the second anomaly fired, did anyone confuse the symptoms with a continuation of the first? 3. Was the FD's prioritisation clean (transponder first, then wheel work)? 4. What's the cognitive load lesson here? Multi-anomaly is where teams typically break. 5. Mission Manager: how was the cohort customer bulletin worded when *two* things went wrong in one shift?
